

Continuous distributions for the claim costs

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References

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Continuous distributions

- We need to use continuous distributions for the modelization of the claim costs in non-life insurance (motor, household policies). Some well known continuous distributions that can be used:
 - Lognormal
 - Pareto
 - Exponential
 - Gamma
 - Inverse Gaussian
- In some cases, it would be usefull to use extreme value distributions (such as the Weibull or Gumbel).

Lognormal

- It is widely used for modelling the costs of claims. A variable X is lognormally distributed $X \sim LN(\mu, \sigma^2)$ if its logarithm is normal, $\log(X) \sim N(\mu, \sigma^2)$. Its distribution function and density functions are:

$$F(X) = \Phi\left(\frac{\log x - \mu}{\sigma}\right), x > 0$$

$$f(X) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{\log x - \mu}{\sigma}\right)^2\right), x > 0$$

where $\Phi(\cdot)$ represents the standard Normal distribution function.

Lognormal

- **Moments:**

$$E(X) = \exp\left(\mu + \frac{\sigma^2}{2}\right)$$

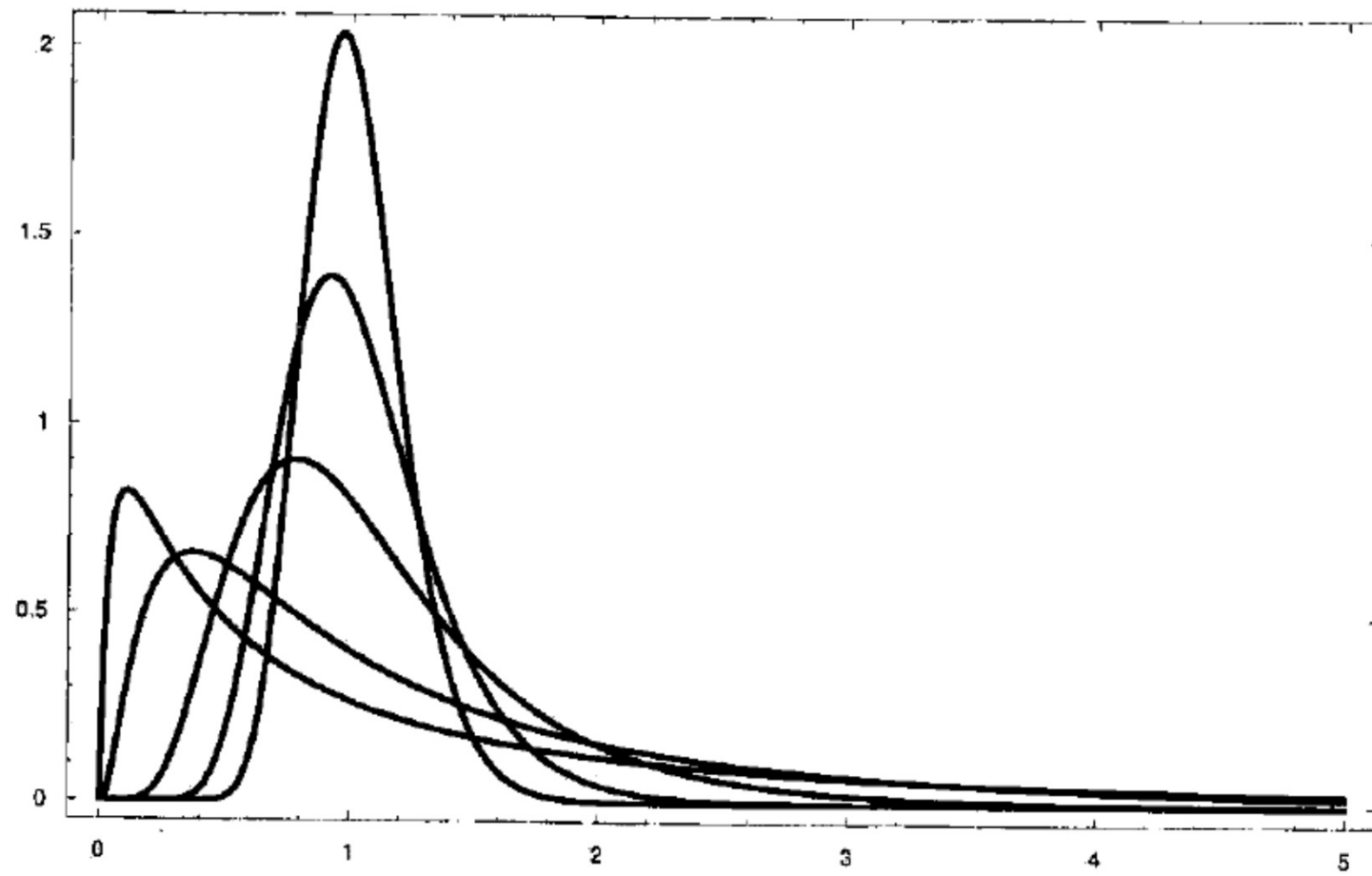
$$\text{Var}(X) = \exp(2\mu + \sigma^2)[\exp(\sigma^2) - 1]$$

$$\gamma_1 = \omega^3 + 3\omega > 0$$

$$\gamma_2 = \omega^8 + 6\omega^6 + 15\omega^4 + 16\omega^2 > 0$$

with $\omega = \sqrt{\exp(\sigma^2) - 1}$, therefore, it is an asymmetric and leptokurtic distribution.

Lognormal



Lognormal

- **Estimation:** the likelihood function based on a sample of size n is:

$$L(\mu, \sigma^2; X) = \sigma^{-n} (2\pi)^{-n/2} \left(\prod_{i=1}^n X_i^{-1} \right) \exp \left\{ -\frac{1}{2} \sum_{i=1}^n \left(\frac{\log X_i - \mu}{\sigma} \right)^2 \right\}$$

and the estimators are:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n \log X_i$$

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (\log X_i - \hat{\mu})^2$$

Pareto

- The Pareto distribution has been widely used in actuarial science, specially for modelling the tail of the distribution of costs, because it converges to 0 slower than the lognormal. A variable X is Pareto I distributed $X \sim \text{PaI}(\alpha, \sigma)$ if the distribution function is :

$$F(X) = \begin{cases} 1 - \left(\frac{x}{\sigma}\right)^{-\alpha} & x \geq \sigma > 0 \\ 0 & x < \sigma \end{cases}$$

where $\alpha > 0$ is the shape parameter and σ is the scale parameter. The density function is

$$f(X) = \begin{cases} \frac{\alpha \sigma^\alpha}{x^{\alpha+1}} & x \geq \sigma > 0 \\ 0 & x < \sigma \end{cases}$$

Pareto

- The Pareto distribution has been widely used in actuarial science, specially for modelling the claims costs associated to fire. We assume that the time of a fire T is exponentially distributed $T \sim \varepsilon(1/\alpha)$, therefore $F_T(t) = 1 - \exp(-\alpha t)$ and $E(T) = 1/\alpha$. We also assume that the cost of a fire depends on its duration so that:

$$X = \sigma \exp(kT) \quad \sigma, k > 0$$

Then, the distribution of the cost is Pareto I, $X \sim \text{PaI}(\alpha/k, \sigma)$

Pareto

- **Proof:** We have to calculate the distribution of $X = g(T)$, where $T \sim \varepsilon(1/\alpha)$. In general, we know that we can obtain the probability distribution of a transformed variable, by doing $F_X(x) = F_T(g^{-1}(x))$, therefore:

$$F_T(t) = 1 - \exp(-\alpha t)$$

$$X = g(T) = \sigma \exp(kT)$$

$$g^{-1}(x) = \frac{1}{k} \ln\left(\frac{x}{\sigma}\right)$$

$$F_X(x) = F_T(g^{-1}(x)) = 1 - \exp\left(-\frac{\alpha}{k} \ln\left(\frac{x}{\sigma}\right)\right) = 1 - \left(\frac{x}{\sigma}\right)^{-\alpha/k}$$

which is Pareto I distributed, $X \sim \text{PaI}(\alpha/k, \sigma)$.

Pareto

- **Moments:**

$$E(X) = \frac{\alpha\sigma}{\alpha - 1}, \alpha > 1$$

$$Var(X) = \frac{\alpha\sigma^2}{\alpha(\alpha - 1)^2(\alpha - 2)}, \alpha > 2$$

$$\gamma_1 = 2 \frac{\alpha + 1}{\alpha - 3} \sqrt{1 - \frac{2}{\alpha}}, \alpha > 3$$

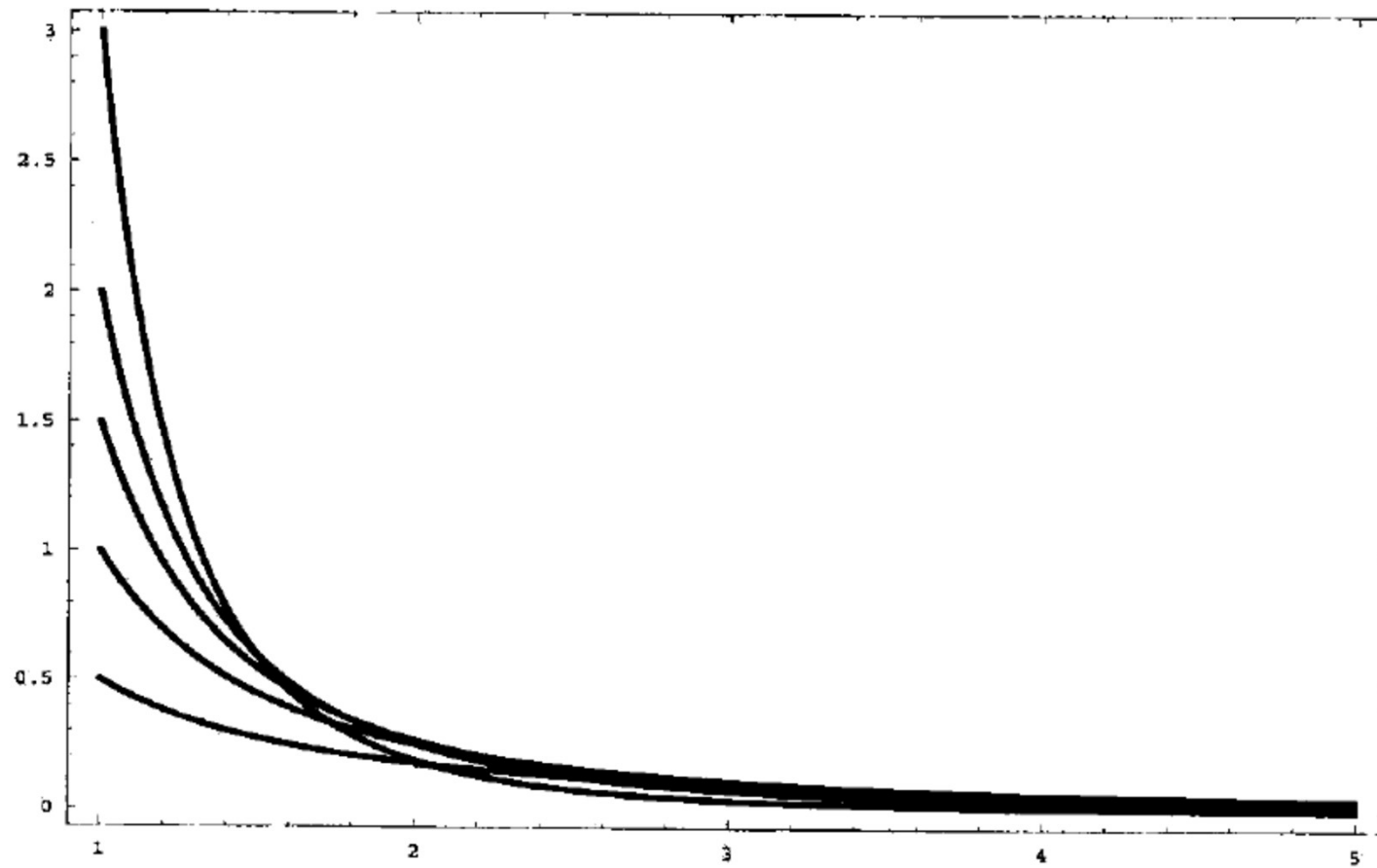
$$\gamma_2 = \frac{6(\alpha^3 + \alpha^2 - 6\alpha - 2)}{\alpha(\alpha - 3)(\alpha - 4)}, \alpha > 4$$

Moreover, $\gamma_1 \rightarrow 2, \gamma_2 \rightarrow 6$ when $\alpha \rightarrow \infty$.

Pareto

Distribution of the cost	Approximated value of α
Earthquake and storms	1
Fire	2
Fire in industries	1.5
Third party liability in motor insurance	2.5

Pareto



Pareto

- **Estimation:** the likelihood function for a sample with a Pareto I distribution is

$$L(\alpha, \sigma; X) = g(T) = \alpha^n \sigma^{n\alpha} \left(\prod_{i=1}^n X_i \right)^{-(\alpha+1)}, \sigma \leq X_{1:n}$$

where $X_{1:n} = \min\{X_1, X_2, \dots, X_n\}$ is the minimum of the sample. The estimators are:

$$\hat{\alpha} = \left(1 - \frac{2}{n} \right) \left[\frac{1}{n} \sum_{i=1}^n \log(X_i/X_{1:n}) \right]^{-1}$$
$$\hat{\sigma} = \left(1 - \frac{\sum_{i=1}^n \log(X_i/X_{1:n})}{n(n-1)} \right) X_{1:n}$$

Exponential

A random variable has an Exponential distribution of parameter σ , $X \sim \varepsilon(\sigma)$ if its density function is

$$f(x) = \begin{cases} \frac{1}{\sigma} e^{-x/\sigma} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

The distribution function is:

$$F(x) = 1 - e^{-\frac{x}{\sigma}} \quad x \geq 0$$

Exponential

Moments:

$$E[X] = \sigma$$

$$Var[X] = \sigma^2$$

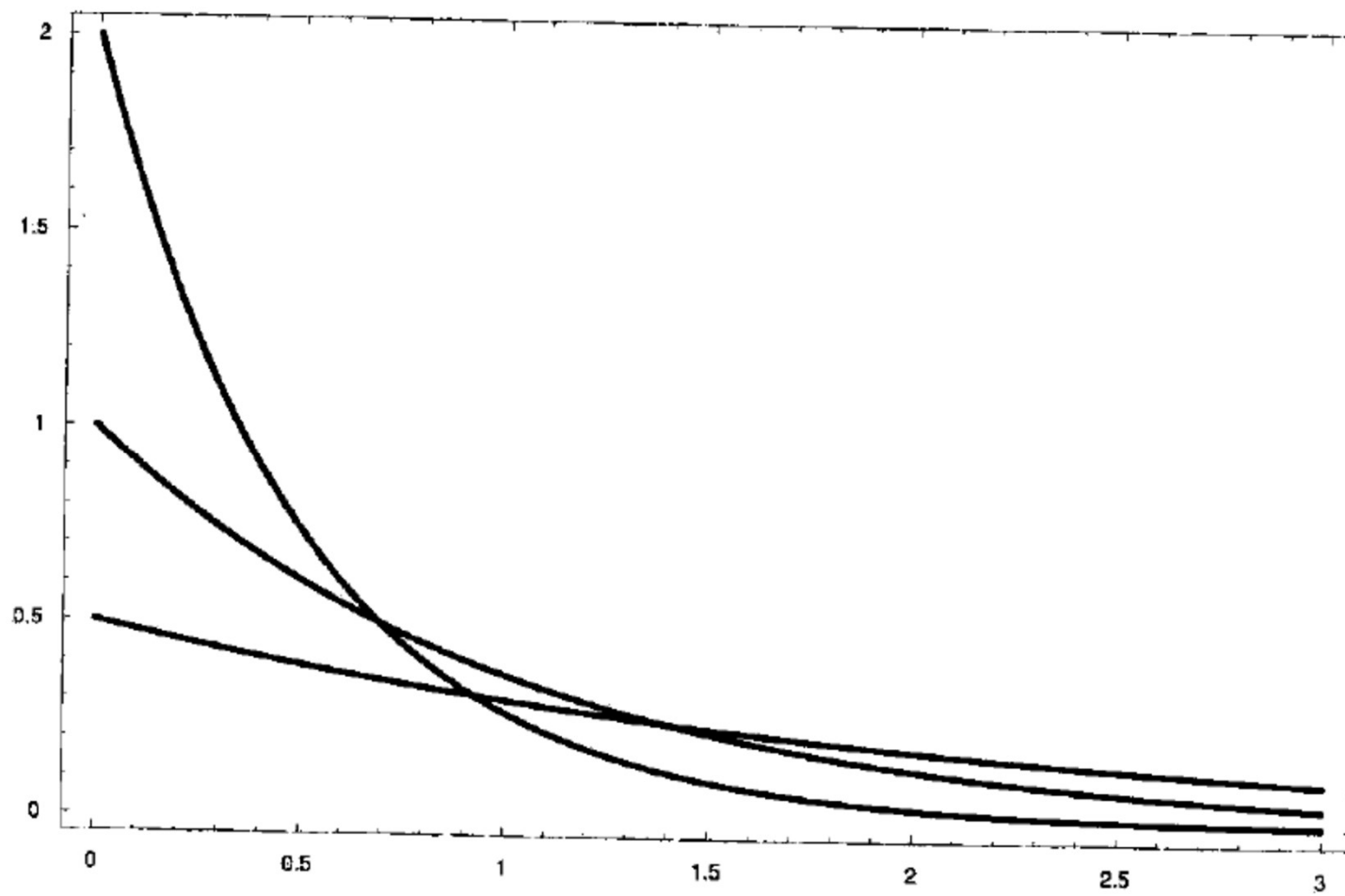
It has positive assymetry ($\gamma_1 = 2$) and it is leptokurtic ($\gamma_2 = 4$).

Moreover, it is the only continuous distribution that fulfills the memoryless property:

$$P(X > s + t | X > s) = P(X > t).$$

Estimation: the ML estimator of σ is the sample mean $\bar{X}$.

Exponential



Gamma

A random variable has a Gamma distribution, $X \sim G(\alpha, \sigma)$ if its density function is

$$f(x) = \begin{cases} \frac{1}{\sigma^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\sigma} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Where α is the shape parameter and σ is the scale.

When $\alpha = 1$ we obtain the Exponential distribution, and if $\alpha \rightarrow \infty$ the Gamma approximates the Normal distribution.

Gamma

Moments:

$$E[X] = \alpha\sigma$$

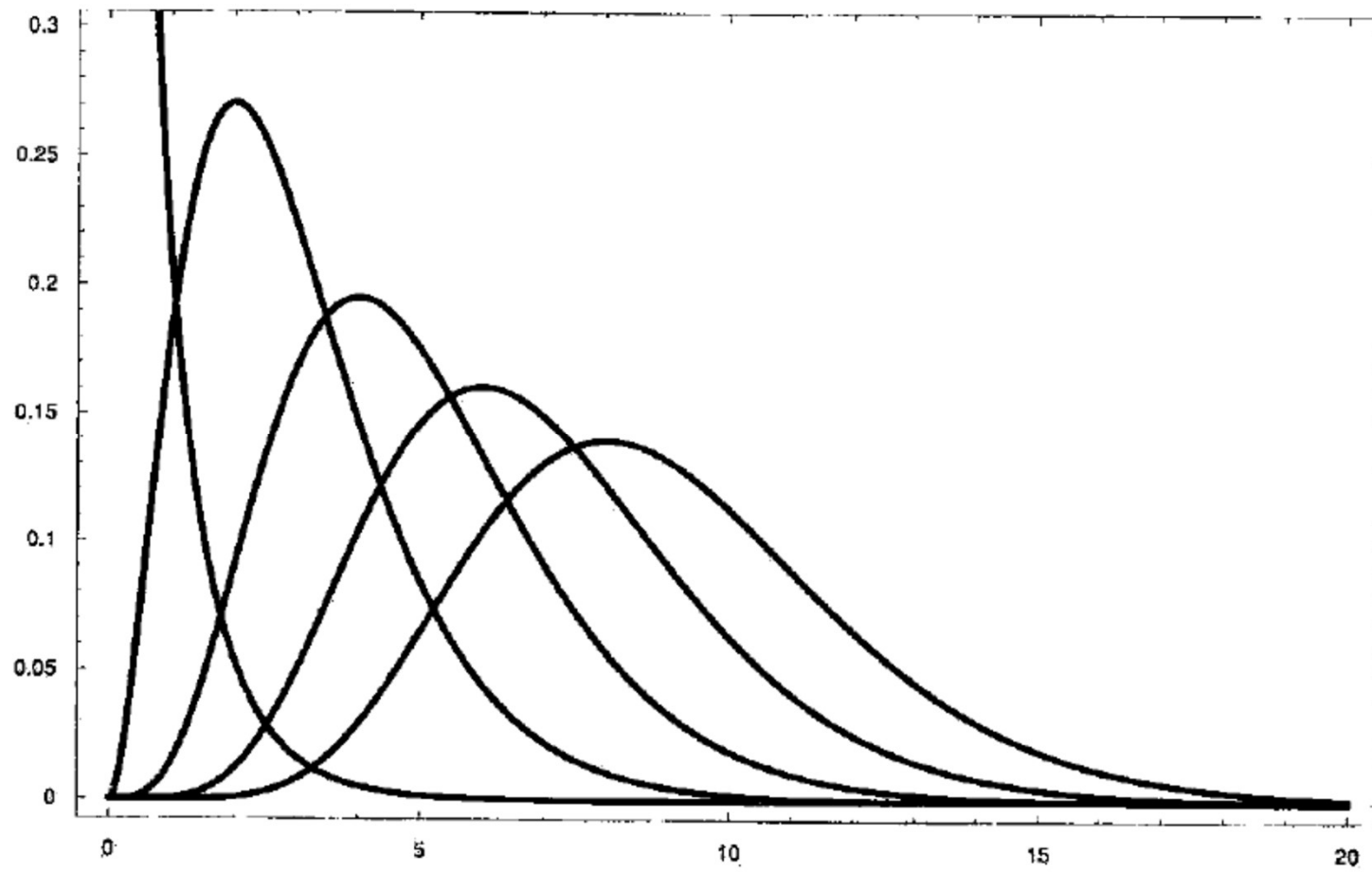
$$Var[X] = \alpha\sigma^2$$

$$\gamma_1 = \frac{2}{\sqrt{\alpha}}$$

$$\gamma_2 = \frac{6}{\alpha}$$

Therefore, it has positive asymmetry and it is leptokurtic.

Gamma



Gamma

Estimation: the likelihood function for a sample of size n equals

$$L(\alpha, \sigma; X) = \frac{1}{\sigma^{n\alpha} [\Gamma(\alpha)]^n} \left(\prod_{i=1}^n X_i \right)^{\alpha-1} \exp \left(- \sum_{i=1}^n X_i / \sigma \right)$$

And the likelihood equations are:

$$\frac{1}{n} \sum_{i=1}^n \log(X_i) = \log \sigma + \psi(\alpha)$$

$$\frac{1}{n} \sum_{i=1}^n X_i = \alpha \sigma$$

where ψ is the digamma function. We get an expression for σ in the second equation and substitute it in the first equation, that only depends on α and it is solved using iterative method (Newton-Raphson). Once $\hat{\alpha}$ is volved, $\hat{\sigma} = \bar{X} / \hat{\alpha}$.

Inverse Gaussian

A random variable has a Inverse Gaussian (or Wald) distribution of parameters μ and λ , $X \sim \text{IG}(\mu, \lambda)$ if its density function is

$$f(x) = \sqrt{\frac{\lambda}{2\pi x^3}} \exp \left\{ -\frac{\lambda}{2\mu^2 x} (x - \mu)^2 \right\}, x > 0, \mu > 0, \lambda > 0$$

The distribution function can be expressed in terms of the Standard Normal distribution function $\Phi(\cdot)$:

$$F(x) = \Phi \left(\sqrt{\frac{\lambda}{x}} \left(\frac{x}{\mu} - 1 \right) \right) + e^{2\lambda/\mu} \Phi \left(-\sqrt{\frac{\lambda}{x}} \left(\frac{x}{\mu} - 1 \right) \right), x > 0, \mu > 0, \lambda > 0$$

Inverse Gaussian

- **Moments:**

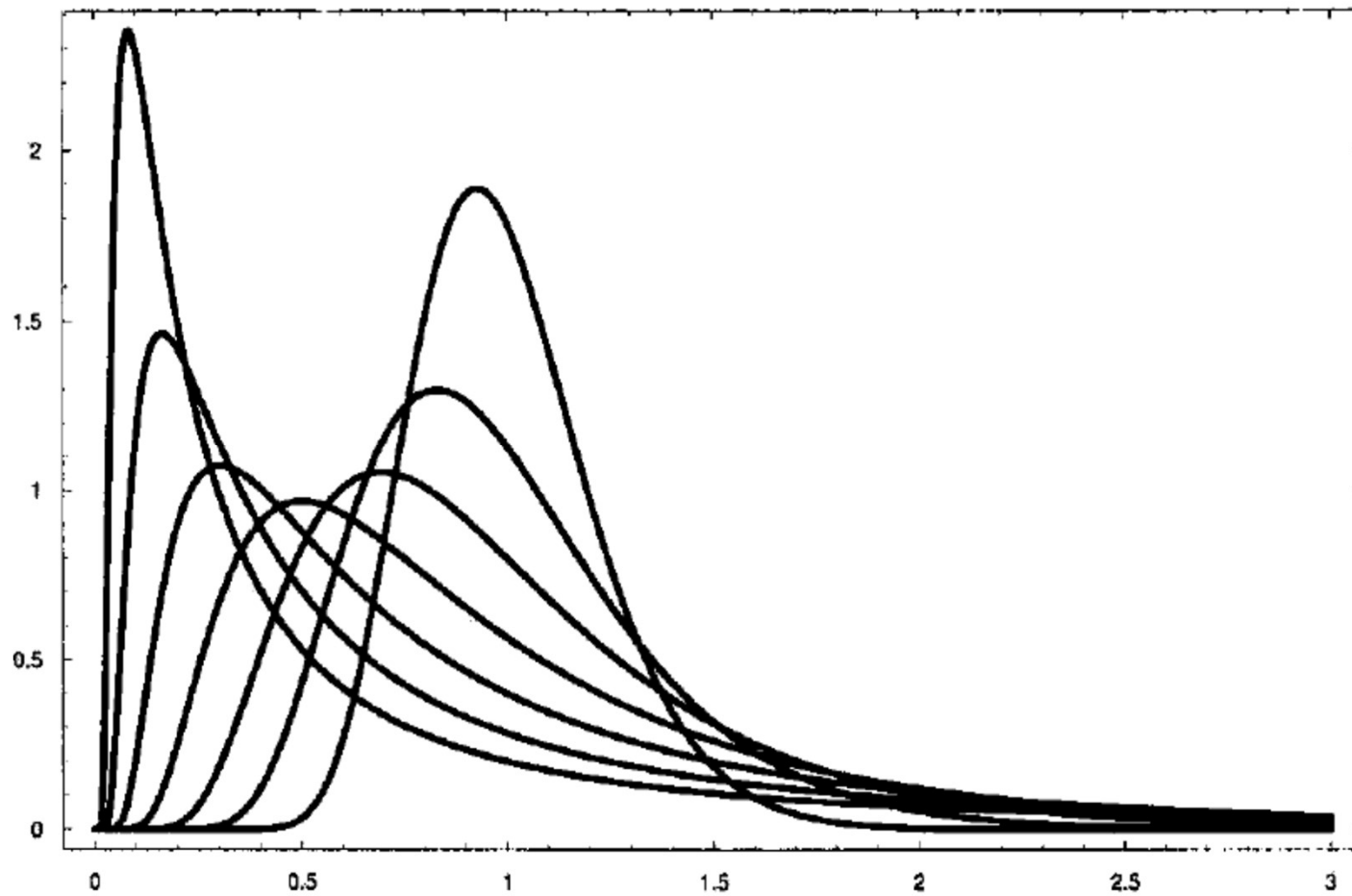
$$E[X] = \mu$$

$$Var[X] = \mu^3 / \lambda$$

moreover, the distribution is asymmetric (positive) and it has one mode equal to

$$\frac{\mu}{2\lambda} \left(\sqrt{4\lambda^2 + 9\mu^2} - 3\mu \right)$$

Inverse Gaussian



Inverse Gaussian

- **Estimation:** for a sample of size n , the MLE of the parameters are:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

$$\hat{\phi} = \frac{1}{\frac{1}{n^2} \sum_{i=1}^n X_i \sum_{i=1}^n X_i^{-1} - 1}$$

It is also possible to use the following parametrization, $IG(\mu, \phi\mu)$. Then, we have that $E[X] = \mu$ and $Var[X] = \frac{\mu^2}{\phi}$.